

Dynamic Macroeconomics II

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A Monetary Model

First Part

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- All models of money have a motive to demand money.
- We will assume that having more money reduces transaction costs.
- I.e. “having more money” in your pocket reduces the cost of acquiring consumption goods.

- Endowment economy with no uncertainty.
- Representative household has one unit of time, which can be allocated to leisure or shopping time: $1 = l_t + s_t$.
- There is a single good, divided between private consumption $\{c_t\}_{t=0}^{\infty}$ and government purchases $\{g_t\}_{t=0}^{\infty}$.
- Utility:

$$\sum_{t=0}^{\infty} \beta^t u(c_t, l_t) \quad 0 < \beta < 1$$

$c_t \geq 0$ and $l_t \geq 0$ are consumption and leisure at time t , respectively.

- $u_c, u_l > 0$, $u_{cc}, u_{ll} < 0$ and $u_{cl} \geq 0$.

- To acquire the consumption good c_t , the household allocates time to shopping s_t .
- We specify amount of shopping time using the function:

$$s_t = H\left(c_t, \frac{m_{t+1}}{p_t}\right)$$

where m_{t+1} is money and p_t is the price level.

- H has the following properties: $H \geq 0$, $H_c, H_{cc} \geq 0$, $H_{m/p} \leq 0$, $H_{m/p, m/p} \geq 0$ and $H_{c, m/p} \leq 0$.
- Example:

$$H = \frac{c_t}{m_{t+1}/p_t} \varepsilon$$

where $\varepsilon > 0$.

- Maximize utility subject to sequential budget constraint and to transaction technology H .
- Sequential budget constraint:

$$c_t + \frac{b_{t+1}}{R_t} + \frac{m_{t+1}}{p_t} = y - \tau_t + b_t + \frac{m_t}{p_t}$$

- m_{t+1} is nominal balances held between periods t and $t + 1$.
- b_t is the real value of one-period government bond holdings and R_t is the real gross rate of return between periods t and $t + 1$.
- Note that b_{t+1} can be positive or negative!
- τ_t is a lump-sum tax and $y > 0$ is the constant value of the endowment.
- Given initial stocks m_0, b_0 , maximization is also subject to $m_{t+1} \geq 0$.

- Consolidate two consecutive budget constraints and arrive at:

$$c_t + \frac{c_{t+1}}{R_t} + \frac{b_{t+2}}{R_t R_{t+1}} + \left(1 - \frac{p_t}{p_{t+1}} \frac{1}{R_t}\right) \frac{m_{t+1}}{p_t} + \frac{m_{t+2}/p_{t+1}}{R_t} = y - \tau_t + b_t + \frac{m_t}{p_t} + \frac{y - \tau_{t+1}}{R_t}$$

- Consider

$$\left(1 - \frac{p_t}{p_{t+1}} \frac{1}{R_t}\right) \frac{m_{t+1}}{p_t}$$

How to ensure budget set (and consumption) is bounded? We need

$$1 - \frac{p_t}{p_{t+1}} \frac{1}{R_t} \geq 0$$

- If

$$1 - \frac{p_t}{p_{t+1}} \frac{1}{R_t} < 0$$

household can make consumption arbitrarily large by choosing an arbitrarily large amount of nominal holdings m_{t+1} .

- Therefore, we need a no-arbitrage condition for bonds and money.
- Note that we can allow for > 0 because we restricted $m_{t+1} \geq 0$
- Define the *real* gross return on money held from t to $t+1$ as $R_{mt} \equiv p_t/p_{t+1}$, that is, the inverse of the inflation *factor*.
- Gross nominal interest rate:

$$1 + i_t \equiv \frac{R_t}{R_{mt}}$$

also known as Fisher's equation.

- Nominal interest rate has a real part (R_t) and an inflationary component ($1/R_{mt} = p_{t+1}/p_t$).
- Back to No-arbitrage condition:

$$1 - \frac{p_t}{p_{t+1}} \frac{1}{R_t} = 1 - \frac{R_{mt}}{R_t} = 1 - \frac{1}{1 + i_t} = \frac{i_t}{1 + i_t} \geq 0.$$

- For this condition to hold we need $i_t \geq 0$, i.e. $R_t \geq R_{mt}$. 
- In words, the gross real return on bonds must be greater than the gross real return on money, which falls with inflation.
- That would be the usual situation, in an economy where impatient consumers ask for a strictly positive real interest rate to save, i.e. $R_t > 1$, and in which there is inflation, i.e. $p_t/p_{t+1} < 1$.

- Lagrangian for household's optimization problem:

$$\begin{aligned}
 L = & \sum_{t=0}^{\infty} \beta^t \left\{ u(c_t, l_t) + \right. \\
 & \lambda_t \left(y - \tau_t + b_t + \frac{m_t}{p_t} - c_t - \frac{b_{t+1}}{R_t} - \frac{m_{t+1}}{p_t} \right) \\
 & \left. + \mu_t \left(1 - l_t - H \left(c_t, \frac{m_{t+1}}{p_t} \right) \right) \right\}
 \end{aligned}$$

- First-order conditions for interior solution:

$$c_t : u_{ct} - \lambda_t - \mu_t H_{ct} = 0 \quad (1)$$

$$l_t : u_{lt} - \mu_t = 0 \quad (2)$$

$$b_{t+1} : -\lambda_t \frac{1}{R_t} + \beta \lambda_{t+1} = 0 \quad (3)$$

$$m_{t+1} : -\lambda_t \frac{1}{p_t} - \mu_t H_{m/pt} \frac{1}{p_t} + \beta \lambda_{t+1} \frac{1}{p_{t+1}} = 0 \quad (4)$$

- From (1) and (2):

$$\lambda_t = u_{ct} - u_{lt} H_{ct} \quad (5)$$

That is, Lagrange multiplier is equal to the marginal utility of consumption less the marginal disutility of shopping time.

- Combine conditions (3) and (5) to obtain:

$$R_t = \frac{1}{\beta} \frac{\lambda_t}{\lambda_{t+1}} = \frac{1}{\beta} \frac{u_{ct} - u_{lt} H_{ct}}{u_{ct+1} - u_{lt+1} H_{ct+1}} \quad (6)$$

Real interest rate depends on discount factor β and is adjusted by shopping time.

- By merging equations (3) and (4) with the definition of R_{mt} , we obtain (**check**):

$$\frac{R_t - R_{mt}}{R_t} \lambda_t = -\mu_t H_{m/p}(t) \quad (7)$$

Interpretation: the cost of holding money ($R_t - R_{mt}$) must be equal to the benefit of an extra marginal amount of real money balances held from t to $t + 1$ ($H_{m/p} \leq 0$).

- Substitute (2) and (5) into (7) to get (**check**):

$$\left(1 - \frac{R_{mt}}{R_t}\right) \left[\frac{u_{ct}}{u_{lt}} - H_c(t) \right] + H_{m/pt} = 0 \quad (8)$$

with u_{ct} y u_{lt} evaluated at $l_t = 1 - H(c_t, m_{t+1}/p_t)$.

- Equation (8) implicitly defines a money demand function, given by:

$$\frac{m_{t+1}}{p_t} = F(c_t, R_{mt}/R_t). \quad (9)$$

- It can be shown that money demand function is increasing in both of its arguments. Need to use the Implicit Function Theorem.
- Therefore the demand for real balances m_{t+1}/p_t is a decreasing function of the nominal interest rate, i.e of the opportunity cost of holding money.
- To see this note that $R_{mt}/R_t = 1/(1 + i_t)$.

- Government finances a stream of purchases $\{g_t\}_{t=0}^{\infty}$ subject to sequential budget constraint:

$$g_t = \tau_t + \frac{B_{t+1}}{R_t} - B_t + \frac{M_{t+1} - M_t}{p_t}$$

taking as given stocks B_0 , and $M_0 > 0$.

- B_t is government indebtedness to private sector and M_t the stock of currency issued as of the beginning of period t .
- Suppose $M_0 = m_0$, and $\{g_t, \tau_t\}_{t=0}^{\infty}$ are exogenous.

Important: This budget constraint comes from consolidating the constraints of the fiscal and monetary branches of a government. See Sargent and Wallace's classic 1981 paper.

- **Price system:** pair of positive sequences $\{R_t, p_t\}_{t=0}^{\infty}$.
- **Equilibrium:** price system, consumption sequence $\{c_t\}_{t=0}^{\infty}$, government debt sequence $\{B_t\}_{t=1}^{\infty}$ and a positive sequence for money supply $\{M_t\}_{t=1}^{\infty}$ such that:
 - 1 Given the price system and taxes, $b_t = B_t$ and $m_t = M_t$ solve household's problem.
 - 2 Government satisfies its sequential budget constraint.
 - 3 Feasibility condition $y = c_t + g_t$ holds.
- $m_t = M_t$ implies equilibrium in the money market.
- $b_t = B_t$ denotes that the government issues debt and households acquire it.

Short run and long run

- Short run: initial period $t = 0$.
- Long run: stationary equilibrium.
- Assume

$$g_t = g \forall t \geq 0$$

$$\tau_t = \tau \forall t \geq 1$$

$$B_t = B \forall t \geq 1.$$

That is, government purchases are always the same, while taxes and debt stock remain constant beginning in $t = 1$.

- We let

$$\tau_0 \neq \tau$$

$$B_0 \neq B.$$

- We can study circumstances in which the economy remains in a stationary equilibrium for $t \geq 1$, but starts from some other position at $t = 0$.

Stationary equilibrium

- Constant: inflation, gross real interest rate, and real variables

$$p_t/p_{t+1} = R_m, \forall t$$

$$R_t = R,$$

$$c_t = c,$$

$$s_t = s,$$

- In a stationary equilibrium, $\lambda_t = \lambda_{t+1}$. Therefore, Euler equation becomes:

$$R_t = \frac{1}{\beta} \frac{\lambda_t}{\lambda_{t+1}} \implies R = \frac{1}{\beta}$$

- Demand for real balances:

$$\frac{m_{t+1}}{p_t} = F(c, R_m/R) \equiv f(R_m) \implies f'(R_m) \geq 0$$

- Notice that c easy to calculate as y, g exogenous: $c = y - g$.

Equilibrium in long run

- Government's budget constraint:

$$g = \tau + \frac{B}{R} - B + M_{t+1}/p_t - M_t/p_t$$

$$g = \tau + \frac{B}{R} - B + f(R_m) - M_t/p_t \frac{p_{t-1}}{p_t}$$

$$g - \tau + B(R - 1)/R = f(R_m)(1 - R_m), \forall t \geq 1$$

- Given (g, τ, B) , we can determine R_m .
- $g - \tau$ is the net-of-interest deficit, the *primary deficit*, and $g - \tau + B(R - 1)/R$ is the gross-of-interest deficit.
- $f(R_m)(1 - R_m)$ is seigniorage revenue from printing currency.
- Inflation tax is $1 - R_m$, increasing in inflation
- Quantity of real balances $f(R_m)$ is base of inflation tax, decreasing in inflation.
- Laffer's curve idea applies to inflation tax: revenue is a concave function of tax rate.

- Government's budget constraint:

$$g = \tau_0 + \frac{B_1}{R} - B_0 + M_1/p_0 - M_0/p_0$$

$$g = \tau_0 + \frac{B}{R} - B_0 + f(R_m) - M_0/p_0$$

$$\frac{M_0}{p_0} = f(R_m) - (g + B_0 - \tau_0) + \frac{B}{R}$$



Initial price level p_0 is determined because M_0 is given in period 0.

Equilibrium determination

- We **first** use equations in the long run to find R_m .
- **Second**, we use the value of R_m to find p_0 .

- Let utility and transaction technology be given by:

$$u(c_t, l_t) = \frac{c_t^{1-\delta}}{1-\delta} + \frac{l_t^{1-\alpha}}{1-\alpha}$$
$$H(c_t, m_{t+1}/p_t) = \frac{c_t}{1 + m_{t+1}/p_t}$$

- Assume $g - \tau + B(R - 1)/R > 0$, i.e. the (fiscal branch of the) government has a deficit.
- Now, let's solve the equilibrium using a graph. Take notes.
- Very important: model has two solutions.
We will work with the "classic" one.

And a result: relation between money growth and inflation.

- In a stationary equilibrium $\frac{m_{t+1}}{p_t} = f(R_m)$ is constant.
- This can only happen if money and the price level grow at exactly the same rate, which is $1/R_m - 1$.
- Inflation is a monetary phenomenon.
- Inflation is determined by the need of the government to finance a fiscal deficit via seigniorage.